Boxfish Abstract

Ostraciidae and Aracanidae are two families of rigid-bodied boxfishes in the subfamily Ostracioidea. The functional morphology of body and fin shape in this taxon is an active area of research due to the opportunity for biomimetic applications to reduce drag, stabilize against roll, and enhance maneuverability in underwater vehicles. To determine whether the observed morphological variation in this taxon evolved independently due to natural selection, it is necessary to build an up-to-date phylogeny of Ostracioidea. The last tree was published over a decade ago. Since then, more sequences have been uploaded to NCBI which can be used to create an updated tree.

I am helping a collaborator studying the form and function of boxfishes to create an updated boxfish tree including species from 31 representative specimens across the superfamily Ostracoidea. To build this tree, I first found published morphological matrices on Aracaniids and Ostraciids. Next, I downloaded available mitochondrial genomes from NCBI and chose a reference sequence off of which to align the rest of the sequences. I generated a preliminary alignment of mitogenomes using MUSCLE and concatenated the sequence matrix with the morphological matrices. After I concatenate a supermatrix of morphological traits, mitochondrial sequences, and nuclear sequences, I plan to run Maximum Likelihood with RAxML and Bayesian Inference with MrBayes to generate and compare the resulting topologies. Lastly, I plan to run a divergence time estimation with BEAST and calibrate the tree using boxfish fossils listed on FishTreeofLife.org which are relatively well-preserved due to the nature of their carapaces.

References:

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